Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 4.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. $6x^2 - x - 15 = 0$

Solve for x.

A $x = -\frac{2}{3}; \frac{3}{5}$ **B** $x = -\frac{5}{3}; \frac{3}{2}$ **C** $x = -1; \frac{5}{2}$ **D** $x = -\frac{3}{2}; \frac{5}{3}$ **E** $x = -\frac{5}{2}; 1$

2. $f(x) = x^2$

$$g(x) = a(x + h)^2$$

Where a and h are constants.

Describe the transformation of the graph of $f(x)$ into the graph of $g(x)$.

A The graph of $f(x)$ is stretched vertically by a factor of a, then shifts h units to the left

B The graph of $f(x)$ is compressed horizontally by a factor of a, then shifts h units to the left

C The graph of $f(x)$ is stretched horizontally by a factor of a, then shifts h units to the left

D The graph of $f(x)$ is compressed vertically by a factor of a, then shifts h units to the left

E The graph of $f(x)$ is stretched vertically by a factor of a, then shifts h units to the right

3. The surface area and volume of a cube have the same value.

Find the volume of the cube.

A 27 cubic units **B** 81 cubic units **C** 144 cubic units **D** 216 cubic units **E** 256 cubic units

4. A coin and a die are tossed.

What is the probability of getting a head and a number greater than 2?

A 1/2 **B** 1/3 **C** 1/4 **D** 1/6 **E** 1/12

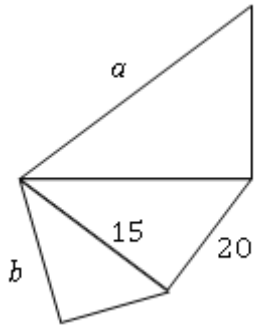
5. Box A has 5 blue marbles, 3 white marbles, and 2 green marbles - all 10 are labeled with 'A'. Box B has 2 blue marbles, 4 white marbles, and 4 green marbles - all 10 are labelled with B. One marble is drawn from Box A and is placed in Box B.

A marble is now drawn from Box B.

What is the probability that the marble drawn from Box B is a white marble labeled with A?

A 3/10 **B** 1/100 **C** 3/100 **D** 1/110 **E** 3/110

6. The figure below shows three similar right triangles (not to scale).



Find $a - b$.

A 92/3 **B** 94/3 **C** 95/3 **D** 97/3 **E** 98/3

7. $4 = \sqrt{(x + \sqrt{(x + \sqrt{(x + \sqrt{(x + \ldots)})})}}$

Solve for x.

A 4 **B** 8 **C** 10 **D** 12 **E** 16

8. A pyramid has a square base with side s and height h. The largest possible right circular cone is fitted inside the pyramid.

Determine the difference in volume.

A $\frac{s^2h\left(1-\frac{\pi}{4}\right)}{3}$ **B** $\frac{s^2h\left(\frac{\pi}{4}-1\right)}{4}$ **C** $\frac{s^2h\left(1-\frac{\pi}{3}\right)}{3}$ **D** $\frac{s^2h\left(1-\frac{\pi}{4}\right)}{4}$ **E** $\frac{s^2h\left(\frac{\pi}{4}-1\right)}{3}$

9. The finals in a basketball tournament is a best-of-7 series wherein the team to win 4 games wins the tournament. After 4 games, Team Tigers won 3 while Team Lions won only 1 game. The probability that Team Lions win a single game is 30%.

What is the probability that Team Lions will win the tournament?

A 2.7% **B** 6.0% **C** 9.0% **D** 12.0% **E** 27%

10. A pattern is designed such that an equilateral triangle is inscribed in a circle, then a smaller circle is inscribed in that triangle, and so on. The radius of the larger circle is R.

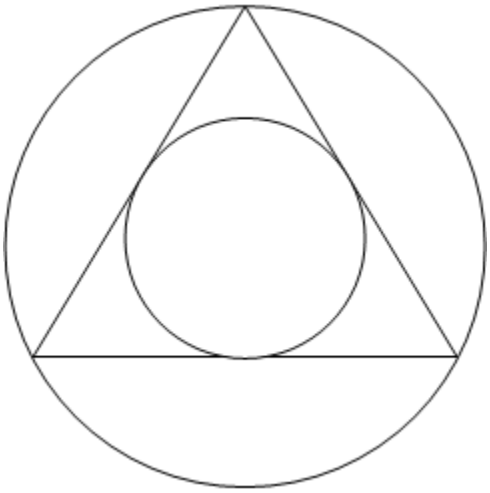


Figure not drawn to scale

Find the length of the side of the third equilateral triangle (if we were to repeat this pattern of drawing an equilateral triangle that fits perfectly inside the circle, then a circle that fits perfectly inside that, etc. For reference, the diagram in the Q contains the 1st triangle, and the 1st and 2nd circles).

A $(R\sqrt{3})/4$ **B** $R/4$ **C** $(R\sqrt{3})/8$ **D** $R/8$ **E** $(R\sqrt{3})/16$

11. 378

392

420

450

Which pair of numbers have a greatest common factor of 28?

A 378 and 392 **B** 392 and 420 **C** 420 and 450 **D** 378 and 450 **E** 378 and 420

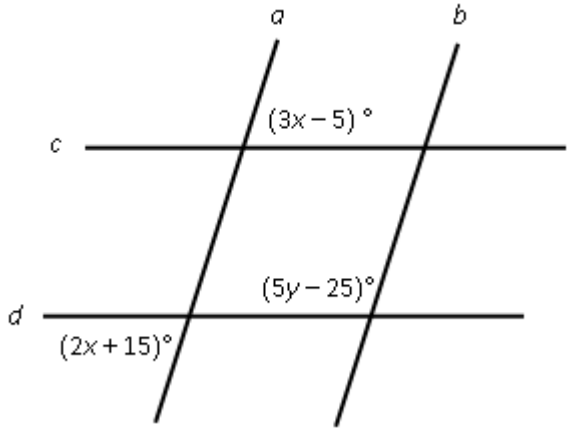
12. Find the values of x and y.

$$2x + 3y = 6$$

$$3x - 2y = -17$$

A $x = 3, y = 4$ **B** $x = -4, y = 3$ **C** $x = -3, y = 4$ **D** $x = 4, y = -3$ **E** $x = 3, y = -4$

13. In the figure, $a \parallel b$ and $c \parallel d$.



Find $x + y$.

A 40 **B** 45 **C** 50 **D** 55 **E** 60

14. $N = 0.181818...$

Express N as a fraction in simplest form.

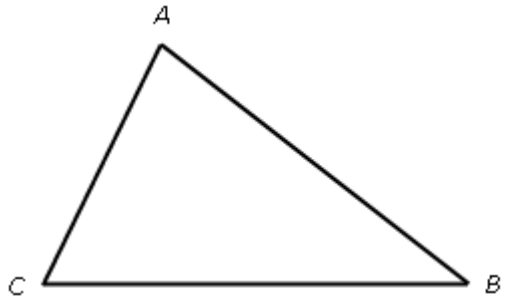
A 2/9 **B** 2/11 **C** 3/11 **D** 3/13 **E** 4/13

15. The points $A(1, -4)$ and $B(3,2)$ are points of a linear function.

Find the x-intercept of the function.

A $(3/7, 0)$ **B** $(0, 7/3)$ **C** $(-7, 0)$ **D** $(0, 3/7)$ **E** $(7/3, 0)$

16. In the triangle shown below, $\angle A = 6x + 6$, $\angle B = 4x - 10$, and $\angle C = 5x + 4$.



What is the difference between the largest and smallest angle?

A 34° **B** 36° **C** 38° **D** 40° **E** 42°

17. The hypotenuse of an isosceles right triangle is $12\ cm$.

Find the area of the triangle.

A $18\ sq.cm$ **B** $24\ sq.cm$ **C** $36\ sq.cm$ **D** $48\ sq.cm$ **E** $72\ sq.cm$

18. The ratio of three numbers is 2:3:5. The smallest number is 16.

Find the sum of the numbers.

A 64 **B** 72 **C** 80 **D** 96 **E** 144

19. $4x^2 - y^2 + 4x + 1$

Factorize the polynomial expression.

A $(2x + y + 1)(2x - y + 1)$

B $(2x - y + 1)(2x - y + 1)$

C $(2x - y - 1)(2x - y - 1)$

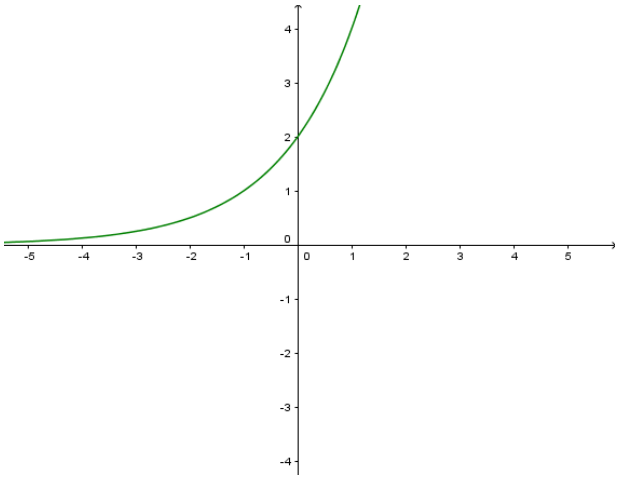
D $(2x + y + 1)(2x + y - 1)$

E $(2x - y - 1)(2x + y + 1)$

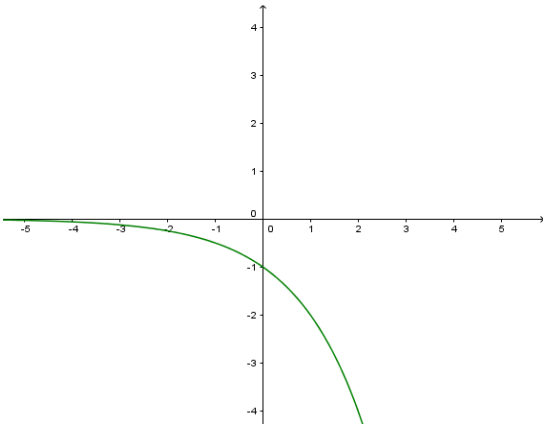
20. $f(x) = -2^{x+1}$

Which of the following is the graph of f(x)?

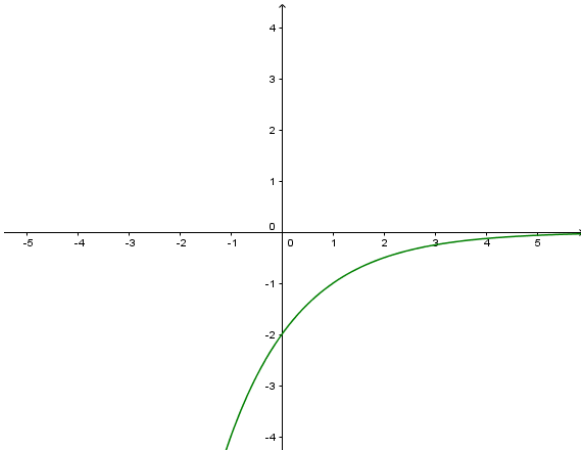
A



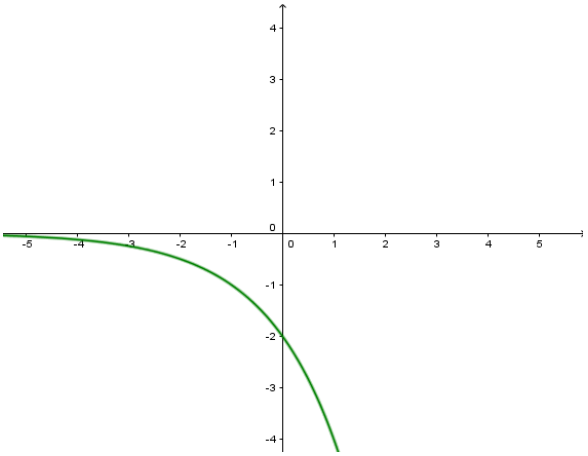
B



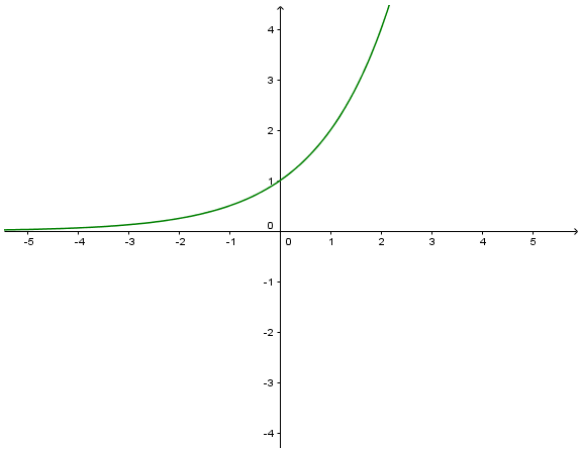
C



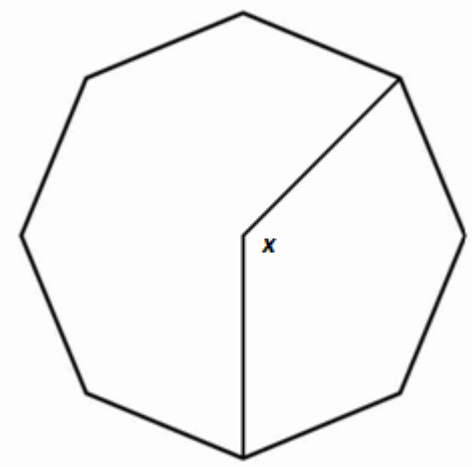
D



E



21. The figure shown is a regular polygon.



What is the measure of angle x?

A 105° **B** 115° **C** 120° **D** 135° **E** 150°

22.

I. 8, 15, 17

II. 7, 24, 25

III. 9, 40, 42

Which of the following are sides of a right triangle?

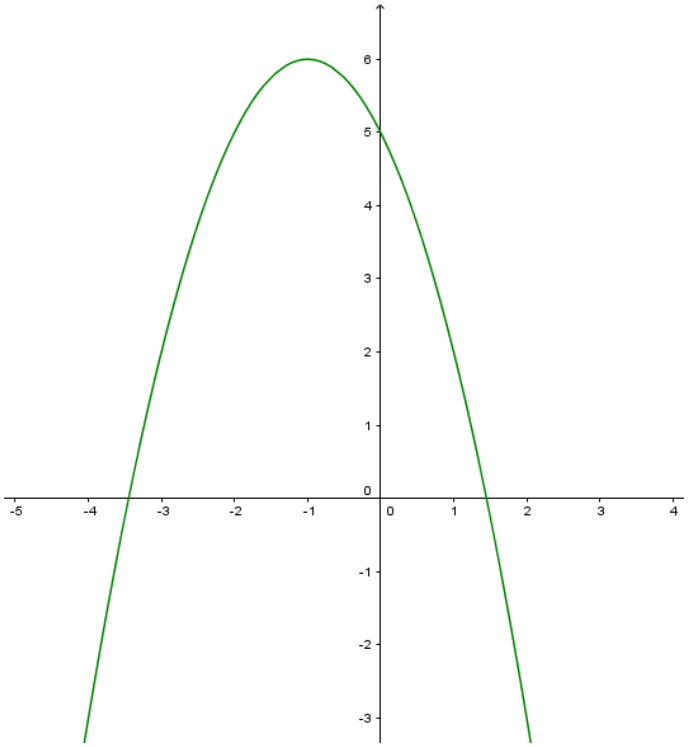
A I only **B** II only **C** I and II only **D** I and III only **E** I, II, and III

23. $2x^2 - 5 = 5x - 2$

Solve for x.

A $x = -\frac{1}{2}; 3$ **B** $x = 1; -\frac{3}{2}$ **C** $x = -\frac{1}{3}; 2$ **D** $x = ½; -3$ **E** $x = -1; \frac{3}{2}$

24. Which of the following is the correct quadratic function?



A $f(x) = x^2 - 2x + 5$

B $f(x) = -x^2 + 2x + 5$

C $f(x) = -x^2 + 2x - 5$

D $f(x) = x^2 - 2x - 5$

E $f(x) = -x^2 - 2x + 5$

25. A multiple-choice test has 3 questions. Answer choices are labeled A, B, and C.

Assuming you knew absolutely nothing about the topic in question and were therefore guessing randomly, what are the chances that you'd get 100%?

A 1/3 **B** 1/6 **C** 1/9 **D** 1/12 **E** 1/27

26. Consider the following set of numbers: {36, 48, 60}

What is the lowest common multiple?

A 360 **B** 480 **C** 540 **D** 720 **E** 960

27. Solve for $x, y,$ and z.

$$x + y + z = 0$$

$$x - y = 1$$

$$y - z = -2$$

A $x = 0, y = -1, z = 1$

B $x = 1, y = -1, z = 0$

C $x = -1, y = 0, z = 1$

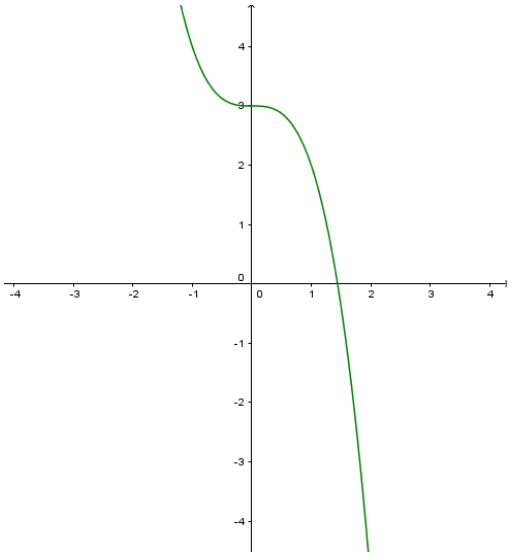
D $x = 0, y = 1, z = -1$

E $x = -1, y = 1, z = 0$

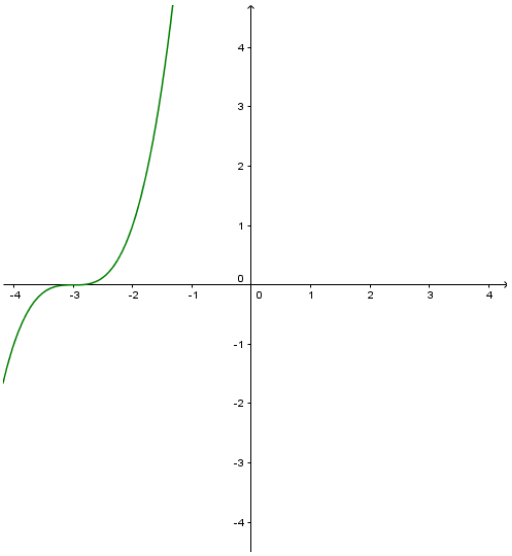
28. $f(x) = x^3 + 3$

Which of the following is the graph of $f(x)$?

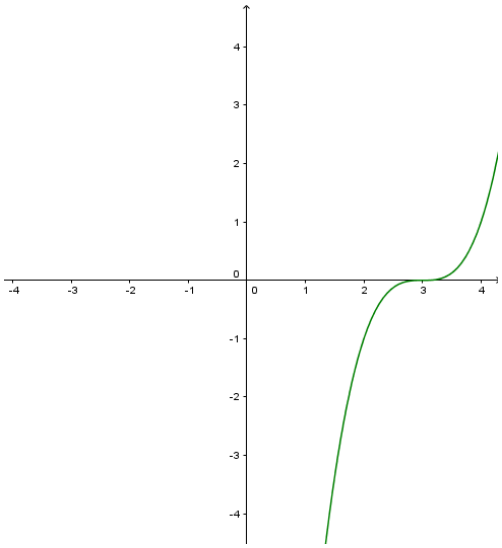
A



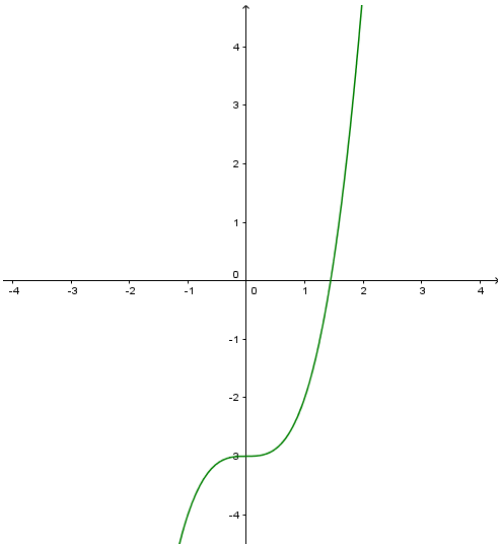
B



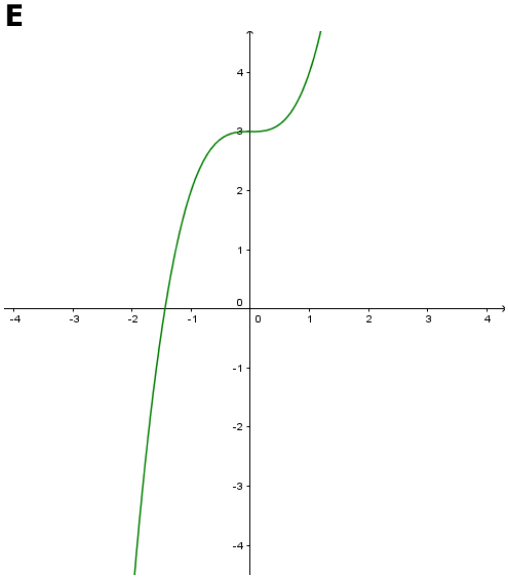
C



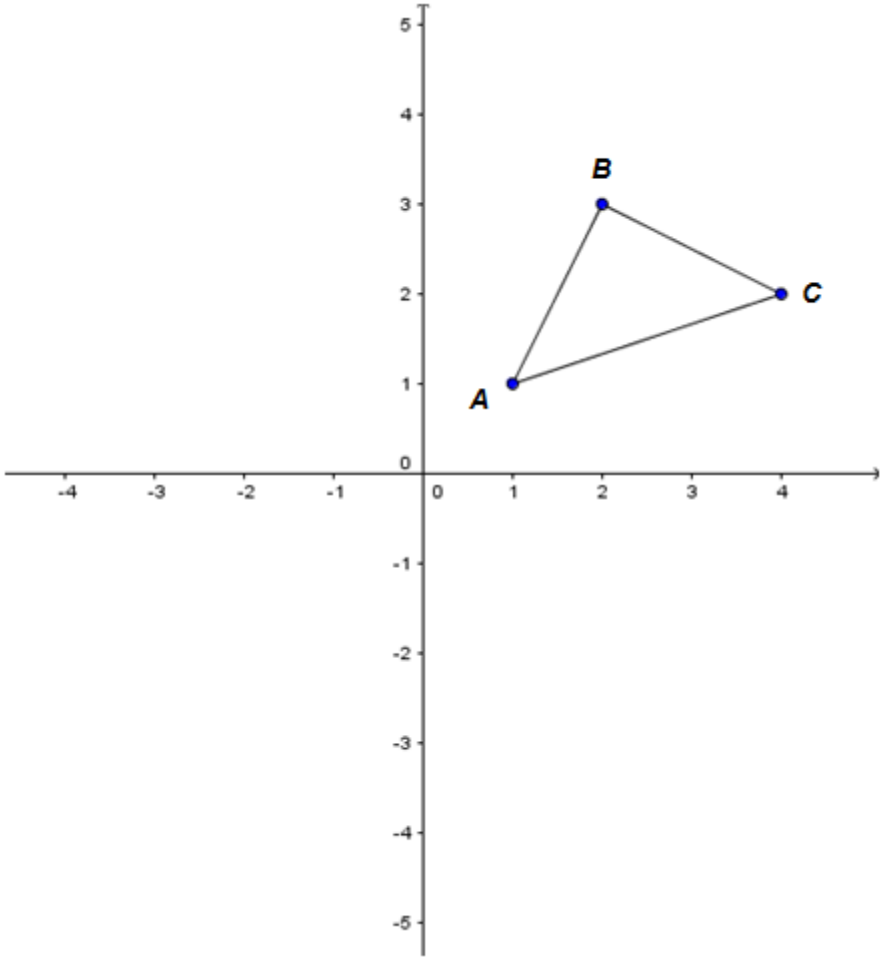
D



E



29. Rotate ΔABC by 90° clockwise, about the origin, and then reflect in the x-axis.



What are the coordinates of the new triangle?

A A(-1,1), B(-3,2), C(-2,4)

B A(-1,-1), B(-3,-2), C(-2,-4)

C A(1,1), B(3,2), C(2,4)

D A(1,-1), B(3,-2), C(2,-4)

E A(-1,-1), B(-2,-3), C(-4,-2)

30. 100 students were surveyed about whether they liked coffee, milk, both or neither. In total, 66 students allegedly like coffee, while 32 like milk. 15 students said they didn't like either coffee or milk.

How many students said they liked both?

A 10 **B** 12 **C** 13 **D** 15 **E** 16

Answer key:

№	Correct Answer	Your answer
1	D	
2	A	
3	D	
4	B	
5	E	
6	E	
7	D	
8	A	
9	A	
10	A	
11	B	
12	C	
13	C	
14	B	
15	E	
16	D	
17	C	
18	C	
19	A	
20	D	
21	D	
22	C	
23	A	
24	E	
25	E	
26	D	
27	A	
28	E	
29	C	
30	C	
		Total: _____